

Roll No. _____

Integrated M.C.A. (Semester - VII)
Examination - Dec. - 2025
IMCA - 704T
Design and Analysis of Algorithms

Time Allowed: Three Hours

Maximum Marks: 70
Min. Marks: 28

This question paper contains two sections as under

Section-A

(खण्ड-अ)

Max. Marks-20

This section will have 10 questions, having 02 questions from each unit. Answer in about 20 words. All questions will have equal marks.

(10×02=20)

Section-B

(खण्ड-ब)

Max. Marks-50

This section will have 10 questions, having 02 questions from each unit, one question to be attempted from each unit. Words limit approximately 400-500 words. Questions may have sub divisions. All questions will have equal marks.

(10×05=50)

052

Section - A

2x10=20

1.

- i) Define Algorithm and its properties.
- ii) Explain theta (θ) notation.
- iii) Describe null graph and degree of a Vertex.
- iv) Explain heap.
- v) What is divide - and- conquer? Explain.
- vi) What is spanning tree. Explain.
- vii) Define back tracking and its use.
- viii) Describe travelling salesman problem.
- ix) Explain polynomial time algorithm
- x) Explain hard problems.

1455

Section - B

Attempt one question from each unit.

052

Unit - I

2. ✓ Explain time and space complexity in detail.

OR

Write an algorithm to find the sum of N numbers and calculate time complexity of the algorithm.

Unit - II

3. Explain hashing and its different techniques in detail.

OR

What is BFS? How it differs from DFS. Explain.

Unit - III

4. Suppose $n = 4$, $(p_1, p_2, p_3, p_4) = (100, 10, 15, 27)$ and $(d_1, d_2, d_3, d_4) = (2, 1, 2, 1)$

Find the feasible and then optimal solution for this problem.

OR

Explain the following :

052

- (i) Greedy method.
- (ii) Strassen's matrix multiplication.

Unit - IV

5. Describe 8-Queen's problem and its solution also.

OR

Write short notes :

- (i) Graph Coloring
- (ii) Sum of subsets.

052

Unit - V

6. Explain deterministic, non-deterministic and NP - complete problems with examples.

OR

Describe Cook's theorem.